MATH 118 Homework Assignment 3

Fatih Emre Ogan
230104004090
Section 2

Q18

(1.1) $n=10$

$P(\text{defect}) = 0.02 \Rightarrow P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$

$k = 2$ defects $\quad P(X=2) = \binom{10}{2}(0.02)^2 \cdot (0.98)^8$

$P(X=2) \approx 45 \times 0.0004 \times 0.8507 = 0.0153$

$\boxed{P(X=2) \approx 0.0153}$

(1.2)

at most 1 chip is defective:

$n=10$
$p=0.02$

$P(X \leq 1) = P(X=0) + P(X=1)$

$P(X=0) = \binom{10}{0}(0{,}02)^0 (0{,}98)^{10} \approx 0.8171$

$P(X=1) = \binom{10}{1}(0{,}02)^1 (0{,}98)^9 \approx 0{,}1667$

$P(X \leq 1) \approx 0.8171 + 0{,}1667 = 0.9838$

$\boxed{P(X \leq 1) \approx 0.9838}$

(1.3)

Since n is large and p is small we use Poisson approximation:

$\lambda = np = 100 \cdot 0{,}02 = 2$ so: $X \approx \text{Poisson}(2)$

$P(X=2) = \dfrac{e^{-\lambda}\lambda^2}{2!} = \dfrac{e^{-2} \cdot 2^2}{2!} = 2e^{-2}$, $e^{-2} \approx 0.1353$

$P(X=2) \approx 2 \cdot 0{,}1353 = 0{,}2706$

$\boxed{P(X=2) \approx 0{,}2706}$

(1.4)

Parts (a) and (c) are different settings a: $n=10$, c: $n=100$
so their probabilities shouldn't be directly compared.
To compare accuracy, compare Poisson to exact binomial
for $n=100$, $p: 0,02$.

Exact binomial: 0,27341

Poisson : 0,27067

Difference: 0.00274 about 1.00% relative error

with large n, small p and $\lambda = np = 2$, the Poisson approximation is quite accurate, slightly underestimating the true probability.

Q2:

$\alpha = 3$ and $\beta = 500$ $\Gamma(3) = (3-1)! = 2$

(2.1) $X \sim$ Gamma $(\alpha = 3, \beta = 500)$

$$PDF = f(x) = \frac{1}{\Gamma(\alpha)\beta^{\alpha}} x^{\alpha-1} e^{-x/\beta}, \quad x \geq 0$$

$$\boxed{f(x) = \frac{1}{2\cdot(500)^3} x^2 e^{-x/500} \quad x \geq 0}$$

Mean: $\boxed{\text{Mean} = \alpha\beta = 3 \cdot 500 = 1500 \text{ hours}}$

Variance: $\boxed{\text{Variance} = \alpha\beta^2 = 3 \cdot 500^2 = 750000}$

(2.2) $X \sim$ Gamma $(\alpha = 3, \beta = 500)$

$P(X < 1000) = ?$

CDF: $F(x) = 1 - e^{-x/\beta}\left(1 + \frac{x}{\beta} + \frac{(x/\beta)^2}{2}\right)$

$x = 1000$, $\beta = 500$

$\frac{x}{\beta} = 2 \Rightarrow F(1000) = 1 - e^{-2}(1 + 2 + 2)$

$F(1000) = 1 - 5e^{-2}$ $(e^{-2} \simeq 0.1353) = 1 - 5(0.1353)$

$$\boxed{F(1000) = 0.3235}$$

2.3 $P(1000 < X < 2000) = ?$

$P(1000 < X < 2000) = F(2000) - F(1000)$

$F(1000) \approx 0.3235$ we calculated before

$F(2000) = 1 - e^{-4}\left(1 + 4 + \frac{4^2}{2}\right)$

$F(2000) = 1 - e^{-4}(13)$ and $e^{-4} \approx 0.0183$

$F(2000) = 1 - 13(0.0183) = 0,7621$

$\boxed{P(1000 < X < 2000) = 0.7621 - 0.3235 = 0.4386}$

Q3: 3.1

$X \sim N(\mu = 10.0, \sigma = 0.05)$

$P(9.95 < X < 10.05) = ?$

Before convert to standart normal z-rate:

$$z = \frac{X - \mu}{\sigma}$$

lower limit: $z_1 = \dfrac{9.95 - 10.0}{0.05} = -1$ upper limit: $z_2 = \dfrac{10.05 - 10}{0.05} = 1$

So:

$P(9,95 < X < 10.05) = P(-1 < z < 1)$

$z(1.00) = 0.8413$, $z(-1.00) = 0.1587$

$P(-1 < z < 1) = z(1.00) - z(-1.00) = 0.8413 - 0.1587 = 0,6826$

$\boxed{P(-1 < z < 1) \approx 0,6826}$

3.2 $P(z < z_{0.025}) = 0.025$

$z_{0.025} = -1,96$

$x = \mu + z\sigma$

$x = 10.0 + (-1.96)(0.05)$

$\boxed{x = 10.0 - 0,098 = 9902 \text{ mm}}$

3.3

$\mu = 10.0$ mm $\quad \sigma = 0.05$

$\pm 0,10 \Rightarrow 10.0 - 0.10 = 9,90$ and $10.0 + 0,1 = 10.10$

$$P(X < 9,90) + P(X > 10.10)$$

$$z_1 = \frac{9.90 - 10.0}{0.05} = -2$$

$$z_2 = \frac{10.10 - 10.0}{0.05} = 2$$

$P(X < 9.90) + P(X > 10.10) = P(z < -2) + P(z > 2)$

$P(z < 2) = 0,9772 \Rightarrow P(z > 2) = 1 - P(z < 2)$

$P(z > 2) = 1 - 0.9772 = 0,0228$

Because of symmetry:

$P(z < -2) = P(z > 2) = 0,0228$

$P(|Z| > 2) = P(z < -2) + P(z > 2) = 0,0456 \approx 4.56\%$

3.4 $\quad X \sim N(\mu, \sigma)$ (Normal dist)

convert to Z:

$$Z = \frac{X - \mu}{\sigma}$$

This transforms X into a distribution with mean 0 and standart deviation 1.

Probabilities can be obtaned directly from the z-table which list values of:

$$P(Z < z)$$

so z allows us to use one universal table instead of seperate tables for every different mean and standart deviation.